Regularisation

Ridge, Lasso and Elastic Net

The problem

- Many features, few samples → huge, unstable weights
- Correlated features make X^TX nearly singular
- We want simpler models that generalise

Three penalties

$$\text{Ridge}: \quad \|y - Xw\|^2 + \alpha \|w\|_2^2$$

$$\text{Lasso}: \quad \frac{1}{2n} \|y - Xw\|^2 + \alpha \|w\|_1$$

- Elastic net mixes both with l1_ratio

Ridge: closed form

```
def fit(self, X, y):
    X, y = np.asarray(X, float), np.asarray(y, float)
    xm, ym = X.mean(0), y.mean()
    Xc = X - xm
    self.w = np.linalg.solve(Xc.T @ Xc + self.alpha * np.eye(X.shape[1]), Xc.T @ (y - ym))
    self.b = ym - xm @ self.w
    return self
```

Lasso: soft thresholding

$$S(z, \gamma) = \text{sign}(z)\max(|z| - \gamma, 0)$$

- No closed form: |w| isn't differentiable at 0
- Coordinate descent: optimise one weight at a time
- Small correlations snap to exactly zero

Coordinate descent

```
def soft_threshold(z, g):
    return np.sign(z) * np.maximum(np.abs(z) - g, 0.0)

# inside ElasticNet.fit
for j in range(d):
    r = yc - Xc @ w + Xc[:, j] * w[j]        # partial residual
    w[j] = soft_threshold(Xc[:, j] @ r / n, l1) / (col_sq[j] + l2)
```

Trade-offs

- Ridge: $O(n \cdot d^2 + d^3)$, keeps all features, handles collinearity
- Lasso: sparse, interpretable; picks one of correlated features
- Elastic net: sparsity + stability for grouped features
- Always scale features first; tune α by cross-validation

Recap

- Penalise weights to reduce variance
- L2 → shrink; L1 → zero out
- Ridge: closed form; Lasso: coordinate descent
- Elastic net blends both